PHY202 Practice Exam 6 on chapter(s): 25, 26

Name:____

These questions assess your understanding of the material from Chapter(s) 25, 26 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. You shine a laser through air at an incident angle of $33.^\circ$ into an unknown material and measure the angle of refraction to be $33.^\circ$. What is the unknown material's index of refraction?
   (Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex2)

   ANSWER: ____________________________

2. Calculate the critical angle for light traveling in a material that is surrounded by air, given that the material has an index of refraction of 1.4.
   (Chapter 25, Section (4) Total Internal Reflection - cp1e_ch25_ex4)

   ANSWER: ____________________________

3. Calculate the power of the eye when viewing an object $7.8\,\mathrm{km}$ away. Take the lens-to-retina distance to be $2.3\,\mathrm{cm}$.
   (Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2a)

   ANSWER: ____________________________

4. Calculate the power of the eye when viewing an object $11.\,\mathrm{cm}$ away. Take the lens-to-retina distance to be $2.4\,\mathrm{cm}$.
   (Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex2b)

   ANSWER: ____________________________

5. Calculate the speed at which light travels in a particular material given that the index of refraction of the material is 1.6.
   (Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex1)

   ANSWER: ____________________________

6. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot $1.6\,\mathrm{cm}$ away from the lens. What is the power of the lens?
   (Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex5)

   ANSWER: ____________________________

7. A clear-glass light bulb is placed $0.45\,\mathrm{m}$ from a thin convex lens that has a focal length of $0.17\,\mathrm{m}$. Calculate the magnification of the produced image.
   (Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6b)

   ANSWER: ____________________________

8. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 8.4° into the material which has an index of refraction of 1.3. Calculate the angle of refraction in the material.
   (Chapter 25, Section (3) The Law of Refraction - cp1e_ch25_ex3)

   ANSWER: ____________________________

9. What is the size of the image on the retina of a $0.57\ \text{mm}$-diameter strand of hair that is held at arm's length ($50.\ \text{cm}$) away? Take the lens-to-retina distance to be $1.4\ \text{cm}$.
(Chapter 26, Section (1) Physics of the Eye - cp1e_ch26_ex1)

ANSWER: ______________________________

10. A scientist at a natural-history museum is holding a specimen $6.8\ \text{cm}$ from a thin convex magnifying glass of focal length $5.1\ \text{cm}$. What magnification is produced?
(Chapter 26, Section (6) Image Formation by Lenses - cp1e_ch25_ex7)

ANSWER: ______________________________

11. A clear-glass light bulb is placed $0.96\ \text{m}$ from a thin convex lens that has a focal length of $0.13\ \text{m}$. Calculate the distance from the lens to the location of the produced image.
(Chapter 25, Section (6) Image Formation by Lenses - cp1e_ch25_ex6a)

ANSWER: ______________________________

12. What power of spectacle lens is needed to correct the vision of a far-sighted person whose near point is $0.83\ \text{m}$, so that they may see an object clearly that is $35.\ \text{cm}$ away? Assume the spectacle lens is held $1.0\ \text{cm}$ away from the eye by eyeglass frames.
(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex4)

ANSWER: ______________________________

13. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point is $22.\ \text{cm}$? Assume the spectacle lens is held $1.2\ \text{cm}$ away from the eye by eyeglass frames.
(Chapter 26, Section (2) Vision Correction - cp1e_ch26_ex3)

ANSWER: ______________________________

cp1e: College Physics (1st Edition) by OpenStax
PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway
tkd: I did not refer to anything while writing this question